AquaNexis — AtomQUEST 2026 Round 2 Progress Report

Track 2: Water Purifier | Institution: UPES Dehradun | Participation: Solo Prototype | Submission Date: 4 November 2026

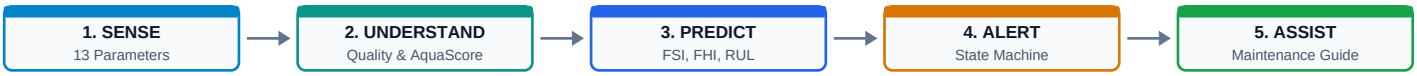
1. Project Positioning & Concise Abstract

Abstract: AquaNexis is a Predictive Water Intelligence System developed for AtomQUEST 2026 Track 2. Domestic reverse osmosis (RO) purifiers currently operate on rigid calendar timers or single-point TDS thresholds, causing either premature filter waste or undetected water quality degradation. AquaNexis delivers an operational software intelligence layer that continuously monitors 13 operating parameters beyond TDS, validates sensor quality, computes a 7-component Filter Stress Index (FSI) and 5-component Filter Health Index (FHI), estimates Remaining Useful Life (RUL) with transparent fallback labeling (MODEL_MISSING), and recommends guided maintenance actions. Operating in DEMO_MODE, the software prototype is 100% verified across 154 unit tests. No drinking-water certification is claimed; physical actuator shutdown is deferred to Round 3.

Field	Specification & Alignment
Primary Exploration	Prognostics of filter failure using data analytics (FSI, FHI, RUL, Failure Risk, Explainability Engine)
Secondary Exploration	Input water quality sensing beyond TDS (TDS, EC, pH, Turbidity, Temp, Flow, Transmembrane Delta-P)
Core Operating Story	Sense (13 parameters) → Understand (Quality Gate) → Predict (FSI/RUL) → Alert (State Machine) → Assist (Maintenance)
Telemetry Provenance	Explicitly tagged: MEASURED (hardware) | SIMULATED (simulator/demo) | CALCULATED (derived) | MODEL_PREDICTED (prognostics)

2. Problem, Proposed Solution & Decision-Flow Architecture

Problem: TDS checks only measure conductive dissolved ions. They cannot detect colloidal turbidity, sediment fouling, or progressive transmembrane pressure drop across RO membranes. **Solution:** AquaNexis establishes a closed-loop intelligence pipeline converting multi-sensor telemetry into explainable prognostic decisions.



3. Completed Modules for Round 2

Module	Status	Verified Technical Capabilities & Grounded Reality
Telemetry Ingestion & Security	Complete	FastAPI ingestion, monotonic sequence replay protection, clock-skew (+/-300s) rejection, JWT claims scoping
Water Quality Engine	Complete	AquaScore (0-100) aggregating TDS, EC, pH, Turbidity, Temp; 1h trend velocity & anomaly z-scores
Filter Stress Index (FSI)	Complete	7-component weighted stress index; dynamic denominator excludes missing sensors without inventing fake data
Filter Health Index (FHI)	Complete	5-component health index (0-100) mapped to maintenance bands: NORMAL, MONITOR, REPLACE_RECOMMENDED
RUL Prognostic Pipeline	Complete	Controlled dataset (controlled-membrane-deg-v1.0), multivariate model (MAE 4.85d), fallback MODEL_MISSING
Explainability & Counterfactuals	Complete	Top 5 contributing degradation drivers (delta-P, flow deviation, volume, age, turbidity) + replacement recovery
Alert State Machine	Complete	Formal lifecycle: DETECTED → OPEN → ACKNOWLEDGED / SNOOZED / RESOLVED / AUTO_CLEARED with evidence drawer
Maintenance Lifecycle	Complete	11-category catalog, replacement guides, sensor calibration records, post-maintenance verification tests
Web Dashboard & Realtime WS	Complete	React 18 + Vite + TypeScript dashboard (localhost:3000), realtime WebSocket stream, scenario switcher

4. Scope Evolution, Reason for Software-First Focus & Sourcing Risks

Strategic Evolution: In Stage 1, a simultaneous physical sensor bench build was conceptualized. For Round 2, the project made a deliberate, strategic pivot to prioritize the software intelligence layer first. This eliminated procurement and bench-rig fabrication risks while ensuring deep prognostic math, sensor validation, and explainability could be thoroughly developed and verified. Firmware architecture in firmware/esp32/ remains preserved as reference for Round 3 physical deployment.

Sourcing / Technical Risk	Stage 1 Plan	Round 2 Implementation	Status & Honest Disclosure
ESP32 & Multi-Sensor Hardware	Physical breadboard rig	Deferred to Round 3; firmware archived	Zero component risk; simulated telemetry in DEMO_MODE
Hydraulic Bench & Actuators	Physical hydraulic pump rig	Deterministic 8-profile Python simulator	Eliminated test-rig leak risks during evaluation window
TimescaleDB Extension	Local PG16 + TimescaleDB	Isolated in-memory repository (DEMO_MODE)	Windows DLL blocker bypassed; CI validates PG16 container

5. Core-Function Evidence & Scenario Results (Verified in DEMO_MODE)

Scenario	Telemetry & Quality State	Intelligence Response (AquaScore / FSI / FHI / RUL / Risk)	Alert & Safety Action Taken
NORMAL	Quality: GOOD, SIMULATED dP=30 kPa, Flow=2.0 L/m	AquaScore=100 (EXCELLENT), FSI=20.0, FHI=86% (NORMAL), RUL=154.8d (MODEL_MISSING), Risk=LOW	0 active alerts; normal background telemetry
FILTER_STRESS	Quality: GOOD, SIMULATED dP=176.9 kPa, Flow=0.96 L/m	AquaScore=93, FSI=67.5 (HIGH), FHI=37% (POOR), RUL=66.6d (MODEL_MISSING), Risk=MEDIUM, Band=REPLACE_RECOMMENDED	Alert: filter_stress_high (CRITICAL); Stage 1 Sediment cartridge replacement recommended
LEAK	Quality: GOOD, SIMULATED leak_detected=True	AquaScore=100, FSI=20.0, FHI=86%, Risk=LOW, Safety Status=CRITICAL_EVENT	Alert: leak_detected (CRITICAL); isolation action logged/recommended; NO physical actuation
SENSOR_FAULT	Quality: BAD, SIMULATED Turbidity out-of-range, missing P_out	AquaScore=90, FSI dynamic denominator active, RUL withheld (N/A, LOW confidence)	Alert: sensor_fault_detected (WARNING); corrupt data prevented from driving false prediction

6. Controlled-Data RUL Machine Learning Validation (Not Field Claims)

To prove mathematical feasibility of prognostic filter degradation, a candidate multivariate regularized regression model was trained and evaluated against a physics-informed dataset. **Disclosure:** These represent controlled-dataset engineering validation results, not physical purifier field claims.

Dataset & Sample Scale	Feature Space	Model Performance Metrics	Operational RUL Runtime Status
600 run-to-failure cycles (480 train / 120 test)	15 engineered physical features (dP, flow deviation, volume, age, etc.)	Candidate Test MAE: 4.85 days R2 = 0.9962 | RMSE = 6.12 days	Deterministic Fallback v1.0 active in UI labeled MODEL_MISSING until field trials

7. Budget & Expenditure Position (Estimate vs Actual)

Category	Stage 1 Budget Estimate	Round 2 Actual Spend	Variance & Financial Justification
Physical Hardware & Sensors	Rs. 5,000 - 10,000	Rs. 0 (Not spent)	Procurement deferred to Round 3 to avoid premature hardware expenditure
Development Tools & Compute	Rs. 0 (Open source)	Rs. 0 (Open source)	Built entirely using open-source FastAPI, React, SQLite/PostgreSQL, and Python
Total Prototype Spend	Rs. 5,000 - 10,000	Rs. 0 (100% within budget)	Entire original hardware budget preserved intact for Round 3 physical testing

8. Safety & Fail-Safe Status

Deterministic Rule Priority: Deterministic rule logic takes priority over predictive output. Leak conditions create a critical safety event and isolation recommendation. Physical automatic shutdown is not claimed until hardware-in-loop validation is complete.

9. Current Limitations & Honest Engineering Disclosures

Disclosures: (1) *No Drinking-Water Certification:* No regulatory certification (e.g., BIS, NSF, WQA) is claimed; AquaNexis is an operational predictive intelligence system; (2) *Simulated Telemetry:* Round 2 telemetry originates from the deterministic simulator (backend/simulator.py) and is explicitly tagged SIMULATED; (3) *RUL Status:* RUL displays MODEL_MISSING fallback in the dashboard until multi-month empirical membrane trials are conducted; (4) *Actuation:* Solenoid valve commands are software recommendations and audit logs only.

10. Automated Verification & Round 3 Engineering Plan

Automated Test Evidence: Backend unit test suite: **154/154 PASS (11 skipped DB-dependent)** in 1.87s across configuration, security, canonical telemetry, ML RUL, hydraulic intelligence, alerts, and maintenance. Frontend checks: **npm run lint PASS** (0 errors, 0 warnings), **npm run format:check PASS**, **npm run typecheck PASS** (0 errors), **npm test PASS (14/14)**, **npm run build PASS** (production bundle produced).

Round 3 Plan: (1) ESP32 physical sensor integration; (2) Sensor calibration against laboratory reference instruments; (3) Real telemetry ingestion pipeline validation; (4) Hardware-in-the-loop (HIL) safety certification for physical Normally Closed solenoid valves; (5) Collection of measured empirical run-to-failure membrane degradation data; (6) Validation and refinement of RUL prediction models using measured field data; (7) Multi-device production deployment with TimescaleDB.